



Evaluation results

There are three levels of importance in pitfalls according to their impact on the ontology:

- **Critical** It is crucial to correct the pitfall. Otherwise, it could affect the ontology consistency, reasoning, applicability, etc.
- **Important** Though not critical for ontology function, it is important to correct this type of pitfall.
- **Minor** It is not really a problem, but by correcting it we will make the ontology nicer.

Pitfalls detected:

Results for P07: Merging different concepts in the same class. 1 case

1 case Minor

A class whose name refers to two or more different concepts is created.

- This pitfall appears in the following elements:

> http://www.cidoc-crm.org/cidoc-crm/E29_Design_or_Procedure

Results for P08: Missing annotations.

16 cases Minor

This pitfall consists in creating an ontology element and failing to provide human readable annotations attached to it. Consequently, ontology elements lack annotation properties that label them (e.g. `rdfs:label`, `lemon:LexicalEntry`, `skos:prefLabel` or `skos:altLabel`) or that define them (e.g. `rdfs:comment` or `dc:description`). This pitfall is related to the guidelines provided in [5].

- The following elements have neither `rdfs:label` or `rdfs:comment` (nor `skos:definition`) defined:

> <http://www.w3.org/ns/prov#Agent>
 > http://iflastandards.info/ns/lrm/lrmoo/F1_Work
 > http://iflastandards.info/ns/lrm/lrmoo/F2_Expression
 > http://iflastandards.info/ns/lrm/lrmoo/F28_Expression_Creation
 > http://www.cidoc-crm.org/cidoc-crm/E39_Actor
 > <http://www.w3.org/2004/02/skos/core#ConceptScheme>
 > <http://www.w3.org/ns/prov#Entity>
 > <http://www.w3.org/2004/02/skos/core#Concept>
 > http://www.cidoc-crm.org/cidoc-crm/E33_Linguistic_Object
 > http://iflastandards.info/ns/lrm/lrmoo/R17_created
 > http://iflastandards.info/ns/lrm/lrmoo/R76_is_derivative_of
 > <http://www.w3.org/ns/prov#wasAttributedTo>
 > http://www.cidoc-crm.org/cidoc-crm/P33_used_specific_technique
 > http://iflastandards.info/ns/lrm/lrmoo/R15_has_fragment
 > http://iflastandards.info/ns/lrm/lrmoo/R3_is_realised_in
 > http://www.cidoc-crm.org/cidoc-crm/P14_carried_out_by

Results for P13: Inverse relationships not explicitly declared.

8 cases Minor

This pitfall appears when any relationship (except for those that are defined as symmetric properties using `owl:SymmetricProperty`) does not have an inverse relationship (`owl:inverseOf`) defined within the ontology.

- This pitfall appears in the following elements:

> http://iflastandards.info/ns/lrm/lrmoo/R17_created
 > http://iflastandards.info/ns/lrm/lrmoo/R76_is_derivative_of
 > <http://www.w3.org/ns/prov#wasAttributedTo>
 > <http://www.w3.org/2004/02/skos/core#inScheme>
 > http://www.cidoc-crm.org/cidoc-crm/P33_used_specific_technique
 > http://iflastandards.info/ns/lrm/lrmoo/R15_has_fragment
 > http://iflastandards.info/ns/lrm/lrmoo/R3_is_realised_in
 > http://www.cidoc-crm.org/cidoc-crm/P14_carried_out_by

Results for P22: Using different naming conventions in the ontology.

Ontology* Minor

The ontology elements are not named following the same convention (for example CamelCase or use of delimiters as "-" or "_"). Some notions about naming conventions are provided in [2].



Suggestions or warnings:

SUGGESTION: symmetric or transitive object properties| 1 case

According to the highest importance level of pitfall found in your ontology the conformace badge suggested is "Minor pitfalls" (see below). You can use the following HTML code to insert the badge within your ontology documentation:



```
<p>
<a href="http://oops.linkeddata.es">
</a>
</p>
```

References

- [1] Aguado-De Cea, G., Montiel-Ponsoda, E., Poveda-Villalón, M., and Giraldo-Pasmin, O.X. (2015).
- [2] Noy, N. F., McGuinness, D. L., et al. (2001).
- [3] Gómez-Pérez, A. (1999).
- [4] Montiel-Ponsoda, E., Vila Suero, D., Villazón-Terrazas, B., Dunsire, G., Escolano Rodríguez, E., Gómez-Pérez, A. (2011).
- [5] Vrandecic, D. (2010).
- [6] Gómez-Pérez, A. (2004).
- [7] Rector, A., Drummond, N., Horridge, M., Rogers, J., Knublauch, H., Stevens, R., Wang, H., and Wroe, C. (2004).
- [8] Hogan, A., Harth, A., Passant, A., Decker, S., and Polleres, A. (2010).
- [9] Archer, P., Goedertier, S., and Loutas, N. (2012).
- [10] Bernes-Lee Tim. (2006).
- [11] Heath, T. and Bizer, C. (2011).
- [12] Vatan, B. (2012).

Enter your ontology to scan:

```
<?xml version="1.0"?>
<rdf:RDF xmlns="https://example.org/desmos/ontology/core/"
  xmlns:base="https://example.org/desmos/ontology/core/"
  xmlns:crm="http://www.cidoc-crm.org/cidoc-crm/"
  xmlns:owl="http://www.w3.org/2002/07/owl#"
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:xml="http://www.w3.org/XML/1998/namespace"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema#">
```

☐ Uncheck this checkbox if you don't want us to keep a copy of your ontology.



How to cite OOPS!

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```
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  title={OOPS! (Ontology Pitfall Scanner!): An On-line Tool for Ontology Evaluation},  
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